VEYLORA

Autonomous Business Process Optimization Platform

Project	Process mining, causal experimentation, and tool-using AI agents for business process optimization
Dataset	BPI Challenge 2017 loan-application event log; current working dataset: 1,592 applications and 71,397 events
Technology	Python, Streamlit, Pandas, NumPy, Plotly, Scikit-learn; PM4Py for offline process analysis; optional local Ollama integration

Abstract

Business processes generate event logs that capture the sequence and timing of activities, but turning these records into actionable operational insight requires methods for process discovery, performance evaluation, and intervention analysis. **Veylora** is a local-first business process intelligence and optimization application that integrates process mining, predictive analytics, causal experimentation, and tool-using AI agents in a single interactive environment. The system analyzes the BPI Challenge 2017 loan-application event log, whose current working analytical dataset contains 1,592 applications and 71,397 recorded events. Its eight workspaces—Overview, Process Explorer, Predictive Evaluation, Opportunity Register, Scenario Lab, Evidence & Governance, AI Assistant, and Autonomous Optimization—support exploration of activity frequencies, event transitions, lifecycle patterns, process variants, case outcomes, and elapsed intervals between recorded events. Predictive models are evaluated against baseline approaches; current results indicate limited predictive performance, so model outputs are treated as exploratory rather than dependable operational forecasts. Veylora also distinguishes historical observational evidence from causal claims: its historical causal assessment reports that causal effects are not identified from the available event log alone. A configurable process simulator creates synthetic treatment and control groups for controlled experiments, allowing hypothetical intervention effects to be evaluated without presenting them as measured historical savings. The tool-using optimization agent selects allow-listed analytical tools in response to detected opportunities, evaluates candidate interventions through synthetic experiments, applies acceptance guardrails, and produces ranked recommendations alongside execution traces and audit records. The implementation includes 23 configured analytical CSV artifacts and an automated test suite with 31 passing tests. Veylora demonstrates an integrated, locally executable workflow for evidence-aware process improvement while emphasizing reproducibility, traceability, and the limitations of observational data and simulation.

Keywords

Process mining; business process optimization; causal inference; predictive analytics; synthetic experimentation; tool-using AI agents; process intelligence; local-first analytics; auditability.

Implementation Snapshot

Analytical artifacts	23 configured CSV artifacts
Application interface	8 interactive Streamlit workspaces
Verification	31 automated tests passed; Python compilation and UI workspace checks completed
Optimization workflow	Allow-listed tool execution, synthetic intervention evaluation, guardrails, ranked recommendations, and audit trace
Execution	Local Windows environment; CPU-compatible workflow

Dataset reference: van Dongen, B. (2017). *BPI Challenge 2017*. 4TU.Centre for Research Data. DOI: 10.4121/uuid:5f3067df-f10b-45da-b98b-86ae4c7a310b.